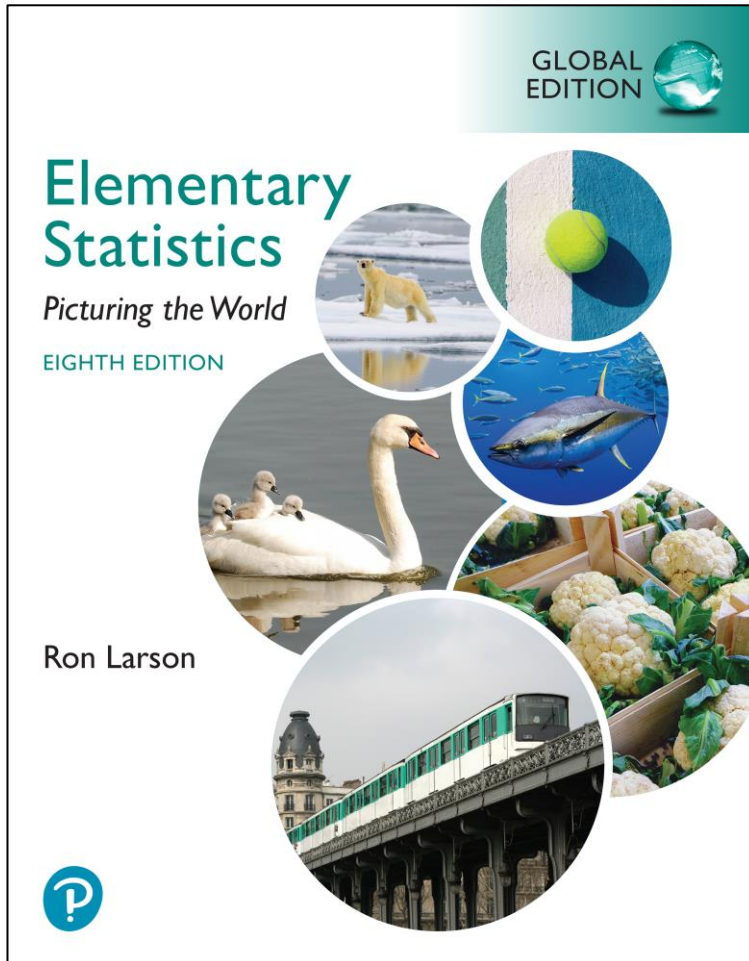


Elementary Statistics

Eighth Edition



Chapter 4

Discrete Probability Distributions

Chapter Outline

4.1 Probability Distributions

4.2 Binomial Distributions

4.3 More Discrete Probability Distributions

Section 4.1 Probability Distributions

Section 4.1 Objectives

1. How to distinguish between discrete random variables and continuous random variables
2. How to construct a discrete probability distribution and its graph and how to determine if a distribution is a probability distribution
3. How to find the mean, variance, and standard deviation of a discrete probability distribution
4. How to find the expected value of a discrete probability distribution

Random Variables (1 of 3)

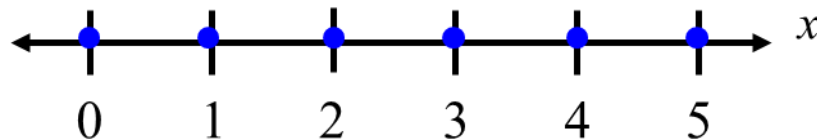
Random Variable

- Represents a numerical value associated with each outcome of a probability distribution.
- Denoted by x
- Examples
 - x = Number of sales calls a salesperson makes in one day.
 - x = Hours spent on sales calls in one day.

Random Variables (2 of 3)

Discrete Random Variable

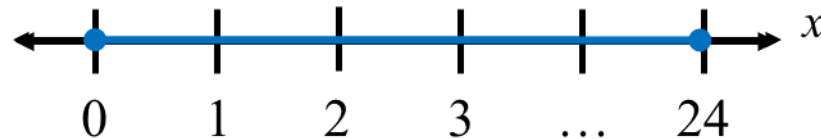
- Has a finite or countable number of possible outcomes that can be listed.
- Example
 - x = Number of sales calls a salesperson makes in one day.



Random Variables (3 of 3)

Continuous Random Variable

- Has an uncountable number of possible outcomes, represented by an interval on the number line.
- Example
 - x = Hours spent on sales calls in one day.



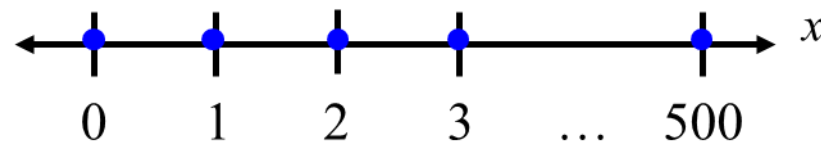
Example: Discrete and Continuous Variables (1 of 2)

Determine whether each random variable x is discrete or continuous. Explain your reasoning.

1. Let x represent the number of Fortune 500 companies that lost money in the previous year.

Solution:

Discrete random variable (The number of companies that lost money in the previous year can be counted.)



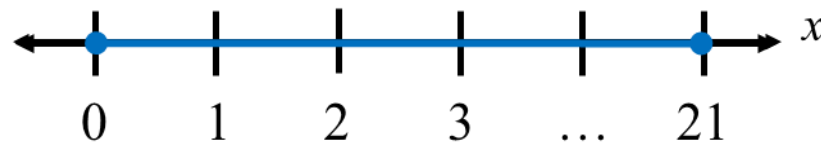
Example: Discrete and Continuous Variables (2 of 2)

Determine whether each random variable x is discrete or continuous. Explain your reasoning.

2. Let x represent the volume of gasoline in a 21-gallon tank.

Solution:

Continuous random variable (The amount of gasoline in the tank can be any volume between 0 gallons and 21 gallons.)



Discrete Probability Distributions

Discrete probability distribution

- Lists each possible value the random variable can assume, together with its probability.
- Must satisfy the following conditions:

In Words	In Symbols
1. The probability of each value of the discrete random variable is between 0 and 1, inclusive.	$0 \leq P(x) \leq 1$
2. The sum of all the probabilities is 1.	$\sum P(x) = 1$

Constructing a Discrete Probability Distribution

Let x be a discrete random variable with possible outcomes $x_1, x_2, \dots, x_n$.

1. Make a frequency distribution for the possible outcomes.
2. Find the sum of the frequencies.
3. Find the probability of each possible outcome by dividing its frequency by the sum of the frequencies.
4. Check that each probability is between 0 and 1, inclusive, and that the sum of all the probabilities is 1.

Example: Constructing and Graphing a Discrete Probability Distribution

An industrial psychologist administered a personality inventory test for passive-aggressive traits to 150 employees. Each individual was given a whole number score from 1 to 5, where 1 is extremely passive and 5 is extremely aggressive. A score of 3 indicated neither trait. The results are shown. Construct a probability distribution for the random variable x . Then graph the distribution using a histogram.

Score, x	Frequency, f
1	24
2	33
3	42
4	30
5	21

Solution: Constructing and Graphing a Discrete Probability Distribution (1 of 3)

- Divide the frequency of each score by the total number of individuals in the study to find the probability for each value of the random variable.

$$P(1) = \frac{24}{150} = 0.16 \quad P(2) = \frac{33}{150} = 0.22 \quad P(3) = \frac{42}{150} = 0.28$$

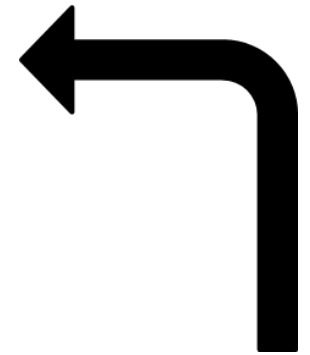
$$P(4) = \frac{30}{150} = 0.20 \quad P(5) = \frac{21}{150} = 0.14$$

- Discrete probability distribution:

<i>x</i>	1	2	3	4	5
<i>P(x)</i>	0.16	0.22	0.28	0.20	0.14

Solution: Constructing and Graphing a Discrete Probability Distribution (2 of 3)

x	1	2	3	4	5
$P(x)$	0.16	0.22	0.28	0.20	0.14



This is a valid discrete probability distribution since

1. Each probability is between 0 and 1, inclusive,

$$0 \leq P(x) \leq 1.$$

2. The sum of the probabilities equals 1,

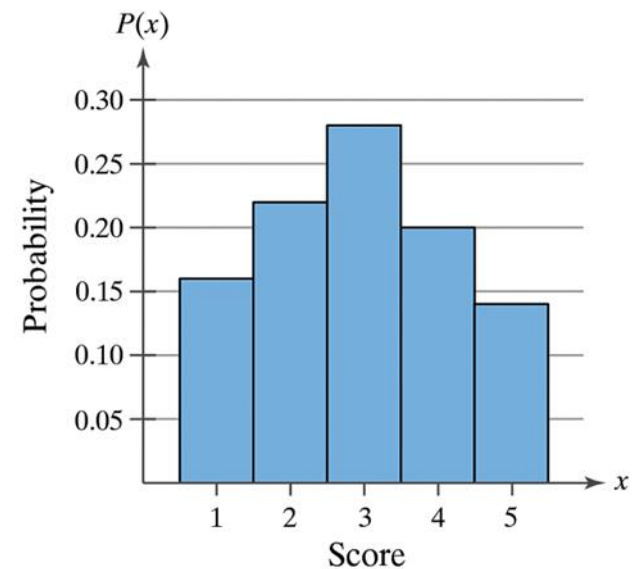
$$\sum P(x) = 0.16 + 0.22 + 0.28 + 0.20 + 0.14 = 1.$$

Solution: Constructing and Graphing a Discrete Probability Distribution (3 of 3)

x	1	2	3	4	5
$P(x)$	0.16	0.22	0.28	0.20	0.14

Because the width of each bar is one, the area of each bar is equal to the probability of a particular outcome. Also, the probability of an event corresponds to the sum of the areas of the outcomes included in the event.

Passive-Aggressive Traits



You can see that the distribution is approximately symmetric.

Example: Verifying a Probability Distribution

Verify that the distribution for the three-day forecast and the number of days of rain is a probability distribution.

Days of Rain, x	0	1	2	3
Probability, $P(x)$	0.216	0.432	0.288	0.064

Solution: Verifying a Probability Distribution

Solution

If the distribution is a probability distribution, then (1) each probability is between 0 and 1, inclusive, and (2) the sum of all the probabilities equals 1.

1. Each probability is between 0 and 1.
2. $\sum P(x) = 0.216 + 0.432 + 0.288 + 0.064 = 1.$

Days of Rain, x	0	1	2	3
Probability, $P(x)$	0.216	0.432	0.288	0.064

Because both conditions are met, the distribution is a probability distribution.

Example: Identifying Probability Distributions (1 of 2)

Determine whether each distribution is a probability distribution. Explain your reasoning.

1.

x	5	6	7	8
$P(x)$	0.28	0.21	0.43	0.15

Solution

- Each probability is between 0 and 1, but the sum of all the probabilities is 1.07, which is greater than 1.
- The sum of all the probabilities in a probability distribution always equals 1. So, this distribution is not a probability distribution.

Example: Identifying Probability Distributions (2 of 2)

Determine whether each distribution is a probability distribution. Explain your reasoning.

2.

x	1	2	3	4
$P(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{5}{4}$	-1

Solution

- The sum of all the probabilities is equal to 1, but $P(3)$ and $P(4)$ are not between 0 and 1.
- Probabilities can never be negative or greater than 1. So, this distribution is not a probability distribution.

Mean

Mean of a discrete probability distribution

- $\mu = \sum xP(x)$
- Each value of x is multiplied by its corresponding probability and the products are added.

Example: Finding the Mean

The probability distribution for the personality inventory test for passive-aggressive traits is given. Find the mean score.

Solution:

x	$P(x)$	$xP(x)$
1	0.16	$1(0.16) = 0.16$
2	0.22	$2(0.22) = 0.44$
3	0.28	$3(0.28) = 0.84$
4	0.20	$4(0.20) = 0.80$
5	0.14	$5(0.14) = 0.70$

$$\mu = \sum xP(x) = 2.94$$

Solution: Finding the Mean

The probability distribution for the personality inventory test for passive-aggressive traits is given. Find the mean score.

Solution: $\mu = \sum xP(x) = 2.94$

Recall that a score of 3 represents an individual who exhibits neither passive nor aggressive traits and the mean is slightly less than 3. So, the mean personality trait is neither extremely passive nor extremely aggressive but is slightly closer to passive.

Variance and Standard Deviation

Variance of a discrete probability distribution

- $\sigma^2 = \sum (x - \mu)^2 P(x)$

Standard deviation of a discrete probability distribution

- $\sigma = \sqrt{\sigma^2} = \sqrt{\sum (x - \mu)^2 P(x)}$

Example: Finding the Variance and Standard Deviation

The probability distribution for the personality inventory test for passive-aggressive traits is given. Find the variance and standard deviation.

Score, x	Probability, $P(x)$
1	0.16
2	0.22
3	0.28
4	0.20
5	0.14

Solution: Finding the Variance and Standard Deviation

Recall $\mu = 2.94$

x	$P(x)$	$x - \mu$	$(x - \mu)^2$	$(x - \mu)^2 P(x)$
1	0.16	$1 - 2.94 = -1.94$	$(-1.94)^2 = 3.7636$	$3.7636(0.16) = 0.60218$
2	0.22	$2 - 2.94 = -0.94$	$(-0.94)^2 = 0.8836$	$0.8836(0.22) = 0.19439$
3	0.28	$3 - 2.94 = 0.06$	$(0.06)^2 = 0.0036$	$0.0036(0.28) = 0.00101$
4	0.20	$4 - 2.94 = 1.06$	$(1.06)^2 = 1.1236$	$1.1236(0.20) = 0.22472$
5	0.14	$5 - 2.94 = 2.06$	$(2.06)^2 = 4.2436$	$4.2436(0.14) = 0.59410$

$$\text{Variance: } \sigma^2 = \sum (x - \mu)^2 P(x) = 1.6164$$

$$\text{Standard Deviation: } \sigma = \sqrt{\sigma^2} = \sqrt{1.6164} \approx 1.27 \approx 1.3$$

Most of the data values differ from the mean by no more than 1.3.

Expected Value

Expected value of a discrete random variable

- Equal to the mean of the random variable.
- $E(x) = \mu = \sum xP(x)$

Example: Finding an Expected Value

At a raffle, 1500 tickets are sold at \$2 each for four prizes of \$500, \$250, \$150, and \$75. You buy one ticket. Find the expected value and interpret its meaning.



Solution: Finding an Expected Value (1 of 2)

- To find the gain for each prize, subtract the price of the ticket from the prize:
 - Your gain for the \$500 prize is $\$500 - \$2 = \$498$
 - Your gain for the \$250 prize is $\$250 - \$2 = \$248$
 - Your gain for the \$150 prize is $\$150 - \$2 = \$148$
 - Your gain for the \$75 prize is $\$75 - \$2 = \$73$
- If you do not win a prize, your gain is $\$0 - \$2 = -\$2$



Solution: Finding an Expected Value (2 of 2)

- Probability distribution for the possible gains (outcomes)



Gain, x	\$498	\$248	\$148	\$73	-\$2
$P(x)$	$\frac{1}{1500}$	$\frac{1}{1500}$	$\frac{1}{1500}$	$\frac{1}{1500}$	$\frac{1496}{1500}$

$$E(x) = \sum xP(x)$$

$$= \$498 \cdot \frac{1}{1500} + \$248 \cdot \frac{1}{1500} + \$148 \cdot \frac{1}{1500} + \$73 \cdot \frac{1}{1500} + (-\$2) \cdot \frac{1496}{1500}$$

$$= -\$1.35$$

You can expect to lose an average of \$1.35 for each ticket you buy.

Example: Finding an Expected Value

A charity organization is selling \$4 raffle tickets as part of a fund-raising program. The first prize is a boat valued at \$3150, and the second prize is a camping tent valued at \$450. The remaining 15 prizes are \$25 gift certificates. The number of tickets sold is 500.

Solution: Finding an Expected Value (1 of 2)

- To find the gain for each prize, subtract the price of the ticket from the prize:
 - You win first prize is $\$3150 - 4 = \3146
 - You win second prize is $\$450 - 4 = \446
 - You win a gift certificate is $\$25 - 4 = \21
- If you do not win a prize, you gain is $\$0 - 4 = -\4

Solution: Finding an Expected Value (2 of 2)

- Probability distribution for the possible gains (outcomes)

Gain (x)	\$3146	\$446	\$21	-\$4
$P(x)$	$\frac{1}{500}$	$\frac{1}{500}$	$\frac{15}{500}$	$\frac{483}{500}$

$$\begin{aligned}E(x) &= \sum xP(x) \\&= \$3146 \cdot \frac{1}{500} + \$446 \cdot \frac{1}{500} + \$21 \cdot \frac{15}{500} + (-\$4) \cdot \frac{483}{500} \\&= \$3.95\end{aligned}$$

- You can expect to gain an average of \$3.95 for each raffle ticket you buy.